Zhaoyi Zhou

(412) 759-4397 • zhaoyiz@andrew.cmu.edu • https://www.linkedin.com/in/zhaoyi-zhou-863577268/

EDUCATION

Carnegie Mellon University August 2024 – May 2029 (Expected)
Ph.D. in Electrical and Computer Engineering Pittsburgh, PA, USA
Advisor: Andrea Zanette

- Main research areas**:** Diffusion language models, LLM agents, reinforcement learning

Carnegie Mellon University August 2024 – Aug 2026
M.S. in Electrical and Computer Engineering Pittsburgh, PA, USA
Earned en route to Ph.D.

- GPA: 4.00/4.00

Tsinghua University September 2020 - June 2024
B.E. in Computer Science and Technology (Yao Class) Beijing, China

- Selected coursework**:** Machine Learning, Theory of Computation, Game Theory, Algorithm Design
- GPA**:** 3.98/4.00

WORK EXPERIENCE

Meta May 2026 – Aug 2026
Research Intern Sunnyvale, CA, USA
Manager: Ayush Jain
Lead the agentic tool selection project.

- Formulate selection of over ten thousand tools as an agentic task.
- Train a small Qwen3 model via RL for tool selection, outperforming Llama-4-Maverick (402B) on ToolBench.
- Demonstrate the transferability of the trained model across retrievers and to new tool corpora domains.

PUBLICATIONS AND PREPRINTS

- Training Small Language Models for Transferable Tool Selection
  Zhaoyi Zhou, Akshayaa Magesh, Jeongyeol Kwon, Artem Kuriksha, Daniel Jiang, Andrea Zanette, Ayush Jain
  Arxiv preprint
- Random Remasking Scales Masked Diffusion Inference
  Zhaoyi Zhou, Zequan Wu, and Andrea Zanette.
  International Conference on Learning Representations 2027. (Under review)
- Variance-Reduced Reinforcement Learning for Large Reasoning Models via James-Stein Baselines.
  Guanning Zeng, **Zhaoyi Zhou**, Daman Arora, and Andrea Zanette.
  International Conference on Machine Learning (ICML) 2026. (26.6% acceptance rate)
- Accelerating Unbiased LLM Evaluation via Synthetic Feedback.
  Zhaoyi Zhou, Yuda Song, and Andrea Zanette.
  International Conference on Machine Learning (ICML) 2025. (26.9% acceptance rate)
- Free from Bellman Completeness: Trajectory Stitching via Model-based Return-conditioned Supervised Learning.
  Zhaoyi Zhou, Chuning Zhu, Runlong Zhou, Qiwen Cui, Abhishek Gupta, and Simon S. Du.
  International Conference on Learning Representations (ICLR) 2024. (31% acceptance rate)
  (ORAL presentation at Foundation Models for Decision Making (FMDM) workshop of NeurIPS 2023.)
- Convergence Rates for Localized Actor-Critic in Networked Markov Potential Games.
  Zhaoyi Zhou, Zaiwei Chen, Yiheng Lin, and Adam Wierman.
  Conference on Uncertainty in Artificial Intelligence (UAI) 2023. (31% acceptance rate)

SELECTED RESEARCH EXPERIENCE

Random Remasking Scales Masked Diffusion Inference Feb 2026 – May 2026
Carnegie Mellon University Pittsburgh, USA

- Propose a novel Markov chain framework of masked diffusion model (MDM) inference, unifying standard decoding and iterative correction.
- Design a reward-free, finetuning-free inference algorithm that scales test-time inference with decoding steps.
- Achieve 98.1% accuracy on hard Sudoku questions, versus 58.9% for the strongest prior baseline.

James-Stein Baseline for Variance-Reduced RLVR in LLM Reasoning Feb 2025 – May 2025
Carnegie Mellon University Pittsburgh, USA

- Propose the novel James-Stein baseline to minimize variance of advantage in RLVR algorithms.
- Theoretically analyze the variance reduction effect of our baseline.
- Empirically demonstrate on GSM8K using Qwen2.5 models that our method achieves lower gradient variance and better test-time performance than RLVR baselines.

Model-Based Return-Conditioned Supervised Learning February 2023 - August 2023
Research Intern. Supervisors: Simon Shaolei Du & Abhishek Gupta Seattle, USA
Led research in offline reinforcement learning (RL).

- Discovered freedom from Bellman completeness requirement as an advantage of return-conditioned supervised learning (RCSL) over dynamic-programming (DP) based offline RL methods.
- Conducted both theoretical and empirical analysis on explicit examples to demonstrate strength of RCSL over DP-based methods in near-deterministic environments given expert dataset.
- Theoretically proved that RCSL cannot do trajectory stitching, a limitation of RCSL observed in previous literature.
- Developed model-based return-conditioned supervised learning (MBRCSL) framework, which enables trajectory stitching of RCSL while avoiding Bellman completeness requirements.
- Tested MBRCSL on offline RL benchmarks, in which MBRCSL outperforms state-of-the-art model-based and model-free offline RL algorithms.

SELECTED HONORS

- Wei Shen and Xuehong Zhang Presidential Fellowship, Carnegie Mellon University, 2026.
- Marija Illic Endowed Fellowship, Carnegie Mellon University, 2026.
- Yao Award, Recognition Prize (presented by Dean Andrew C. Yao of IIIS), Tsinghua University, 2023;
- Scholarship of Academic Excellence, Tsinghua University, 2023 & 2022;
- Scholarship of Scientific Innovation, Tsinghua University, 2023 & 2022;
- Scholarship of Comprehensive Excellence, Tsinghua University, 2021.

PROFESSIONAL ACTIVITIES

Paper Reviewer: ICLR 2025-2027, ICML 2025-2026, NeurIPS 2025-2026.

OTHERS

Software: Python, C/C++, Go, Verilog, MATLAB
Languages: English (Fluent), German (Fluent), Chinese (Native)